CENG-567
REINFORCEMENT LEARNING

Lecture-2

Multi-armed Bandits (Contd.)

Upper Confidence Bounds (UCB), Thompson Sampling

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Contains materials from "Mastering Reinforcement Learning with Python, Enes Bilgin, Packt Publishing, 2020."

Recap

1. Multi-Armed Bandit (MAB) fundamentals

Simplified form of Reinforcement Learning

Single-step decision problem (no future states)

Goal: maximize immediate reward

2. Exploration–Exploitation Trade-Off

Exploration: try new actions to gather information

Exploitation: use known best actions to gain reward

Key challenge: balancing both efficiently

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Recap

3. Exploration strategies

A/B/n Testing:

Fixed test phase, static probabilities
Simple but inefficient and non-adaptive

ε-Greedy:

With prob. (1–ε): exploit best action
With prob. (ε): explore randomly
Better adaptability, still somewhat static

Key takeaway:
Finding the right balance between **exploration** and **exploitation** is essential for optimal decision-making in MAB problems.

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Action Selection Using Upper Confidence Bounds (UCB)

Key Idea:
Upper Confidence Bounds (UCB) is a principled method to balance **exploration** and **exploitation**.

Concept:
At each time step, choose the action with the **highest potential reward**, defined as:

Potential = Estimated Value + Uncertainty

An action may be selected because:

Its estimated reward is high, **or**

It is **less explored**, hence more uncertain, **or both**

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The UCB Formula

We select the action to take at time t that using the following formula:

$$A_t \triangleq \arg\max_a \left[Q_t(a) + c\sqrt{\frac{\ln t}{N_t(a)}} \right]$$

Where:

$Q_t(a)$:estimated action value

$N_t(a)$:number of times action a was selected

t: current time step

c: hyperparameter controlling exploration strength

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Understanding the Formula

Breakdown:

$Q_t(a)$:**exploitation term** → favors actions with high average reward

$\sqrt{\frac{\ln t}{N_t(a)}}$:**exploration term** → favors uncertain actions

As $N_t(a)$ → ↑uncertainty ↓

As t↑ → overall uncertainty ↑ slightly → encourages continued exploration

Tunable parameter:

$\boldsymbol{c}$ adjusts how aggressively uncertainty is explored

Must be tuned carefully; **too low** → under-exploration, **too high** → over-exploration

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Online advertising scenario with UCB

- See '**Multi-armed bandits.ipynb**' file in the Teams course page.

Setup:

Define ads and initialize parameters

At each impression:

Compute the UCB for each ad
Select the ad with the highest UCB
Observe reward (click / no click)
Update counts and estimates

If an action has not been selected yet, it has the highest UCB.

Action Selection Logic

Key behavior of UCB:

Unseen ads → highest uncertainty → selected early

Well-tested ads → lower uncertainty → less exploration

Exploration automatically decreases as more impressions accumulate

Tie-breaking:

If multiple ads have the same UCB → choose randomly

✅ *Efficient exploration without fixed probabilities (unlike ε-greedy).*

Hyperparameter c and Its Effect

c: exploration coefficient

Controls how strongly uncertainty affects decision-making

Low c→ more exploitation

High c→ more exploration

Practical note:

Must be tuned empirically

Performance varies significantly with c

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Example Output

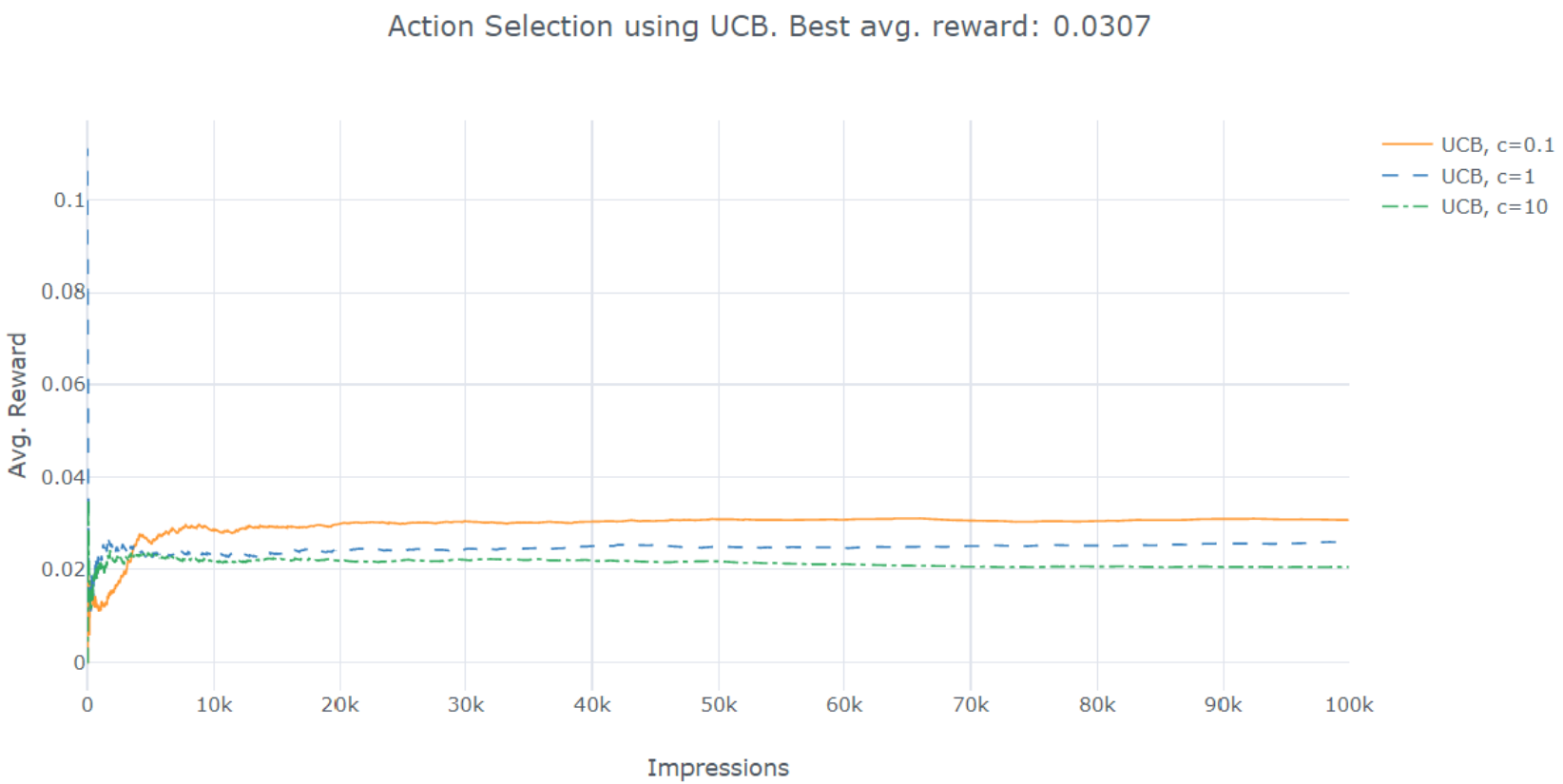

Figure 2.4 – Exploration using UCB

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Meta Observation – Hyperparameter Tuning

Challenge:
Tuning c is **itself a bandit problem**!

Approach:

Try logarithmic scales first: [0.01, 0.1, 1, 10]

Once a good range is found, refine linearly

💡 *In practice, once you find a good c for a specific domain (e.g., online ads), it often generalizes to similar tasks.*

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Advantages and disadvantages of using UCBs

- **UCB is a set-and-forget approach**. It systematically and dynamically allocates the budget to alternatives that need exploration.

- **UCB can be further optimized for dynamic environments, potentially at the expense of introducing additional hyperparameters**. The formula we provided for UCB is a common one, but it can be improved – for example, by using exponential smoothing to calculate the Q values. There are also more effective estimations of the uncertainty component in literature.

- **UCB could be hard to tune**. It is somewhat easier to make the call and say, "I want to explore 10% of the time, and exploit for the rest" for the ε-greedy approach than saying, "I want my to be 0.729" for the UCB approach, especially if you are trying these methods on a brand-new problem.

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UCB in Dynamic Environments: Exponential Smoothing

Goal: Make UCB responsive to **changing reward distributions**.

Key Idea: Update the estimated action value $Q(a)$using **recent rewards more heavily**.

Standard update:

$$Q_{t+1}(a) = Q_t(a) + \frac{1}{N(a)}\left(R_t - Q_t(a)\right)$$

Exponential smoothing update:

$$Q_{t+1}(a) = (1-\alpha)Q_t(a) + \alpha R_t$$

$\alpha \in (0, 1]$ =smoothing factor (large → fast adaptation, small → smoother estimate)

Selection formula (argmax) **remains the same.**

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Thompson (Posterior) Sampling

- Goal in Multi-Armed Bandit (MAB) problems:
  Estimate the parameter(s) of the reward distribution for each arm (e.g., which ad to display). Measure uncertainty about estimate to guide the exploration strategy.

- This problem fits naturally into the **Bayesian inference** framework.

- Bayesian inference **starts with a prior** probability distribution – an initial idea, for the parameter θ – and **updates this prior distribution as data becomes** available.

- So, the Bayesian approach **treats the parameter as a random variable** given the data.

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Bayesian Inference for Thompson Sampling

Start with a **prior distribution** (initial guess for the parameter).

Update prior with observed **data** → obtain **posterior distribution**.

Bayesian interpretation: Treat the **parameter as a random variable** given the data.

Formula: Posterior: $P(\theta \mid X) = \frac{P(X|\theta)P(\theta)}{P(X)}$

X : Data

$P(\theta)$:prior

$P(X \mid \theta)$:likelihood

$P(\theta \mid X)$:posterior

$P(X)$:evidence

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θ is model-dependent

- **Bernoulli**: $\theta = p$ (probability of success).
- **Normal** with **known variance**: $\theta = \mu$.
- **Normal** with unknown mean & variance: $\theta = (\mu, \sigma^2)$.
- **Poisson**: $\theta = \lambda$ (rate).
- **Logistic regression / contextual bandit**: $\theta = \beta$ (vector of coefficients).

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Thompson Sampling in Online advertising scenario

- Each **ad** corresponds to an **arm**.
- A **click** is a **Bernoulli random variable** (1 = click, 0 = no click).
- Goal: Estimate the **click-through rate (CTR)** = probability of a click.
- CTR $\in$ [0, 1], same logic applies to any **binary-outcome** problem.
- Note that many problems other than online advertising have such a binary outcome. So, this discussion here can be extended to such other cases.

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Initial Beliefs (Prior Distribution)

Initially, we have **no reason** to believe one ad's CTR is higher than another.

Therefore, assume a **uniform prior** for each ad's CTR:

$$\theta_k \sim \text{Uniform}(0,1)$$

or equivalently

$$\theta_k \sim \text{Beta}(1,1)$$ *(we'll discuss beta distribution later)*

Each θ_k represents the **true CTR** of ad k, which we are uncertain about.

As data arrive, we'll update this belief.

Updating Beliefs with Observations

When ad k is displayed and **clicked (1)** → update posterior toward 1.

When ad k is displayed and **not clicked (0)** → update posterior toward 0.

Over time:

- **Expected value** of θ_k shifts based on observed clicks.
- **Variance** shrinks → less uncertainty.

Connection to Exploration vs Exploitation:

High variance → more exploration (less data).

Low variance → more exploitation (well-known ad).

Same principle as **UCB**, but derived via **Bayesian inference**.

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Sampling Step – Choosing the Ad

At each time step:

1. For each ad k, draw one sample from its posterior:
$$\tilde{\theta}_k \sim P(\theta_k \mid X)$$
2. Select the ad with the **highest sampled** $\tilde{\theta}_k$.
3. Observe the outcome (click or not), then **update** that ad's posterior.

Intuition:

Ads with high expected CTR → more likely to win samples (exploitation).

Ads with high uncertainty (wide posterior) → produce diverse samples (exploration).

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Update Rules for the Bernoulli Distribution

Now, let's talk about how we update our beliefs when we observe new data.

A common choice for the **prior** is the **Beta distribution**.

Think about it — our unknown parameter θ, which represents the click-through rate (CTR), can only take values between **0 and 1**. The Beta distribution also lives on this interval, so it's a perfect match.

Update Rules for the Bernoulli Distribution

When we combine this **Beta prior** with a **Bernoulli likelihood** (that is, each ad either gets a click or not), something very convenient happens:

The **posterior** also becomes a **Beta distribution**.
This means after each observation, we simply update the two parameters of the Beta distribution — one for successes (clicks) and one for failures (no-clicks).

This property is called **conjugacy**, and the Beta distribution is known as the **conjugate prior** for the Bernoulli distribution.

Because of this, we can easily perform **sequential updates**:
each time a new click or no-click is observed, we just tweak our Beta parameters and carry on — no complicated recalculations needed.

In short:
We model our uncertainty about CTRs with Beta distributions, update them as new data arrive, and always stay within the same distribution family — a simple yet powerful mechanism behind Thompson Sampling.

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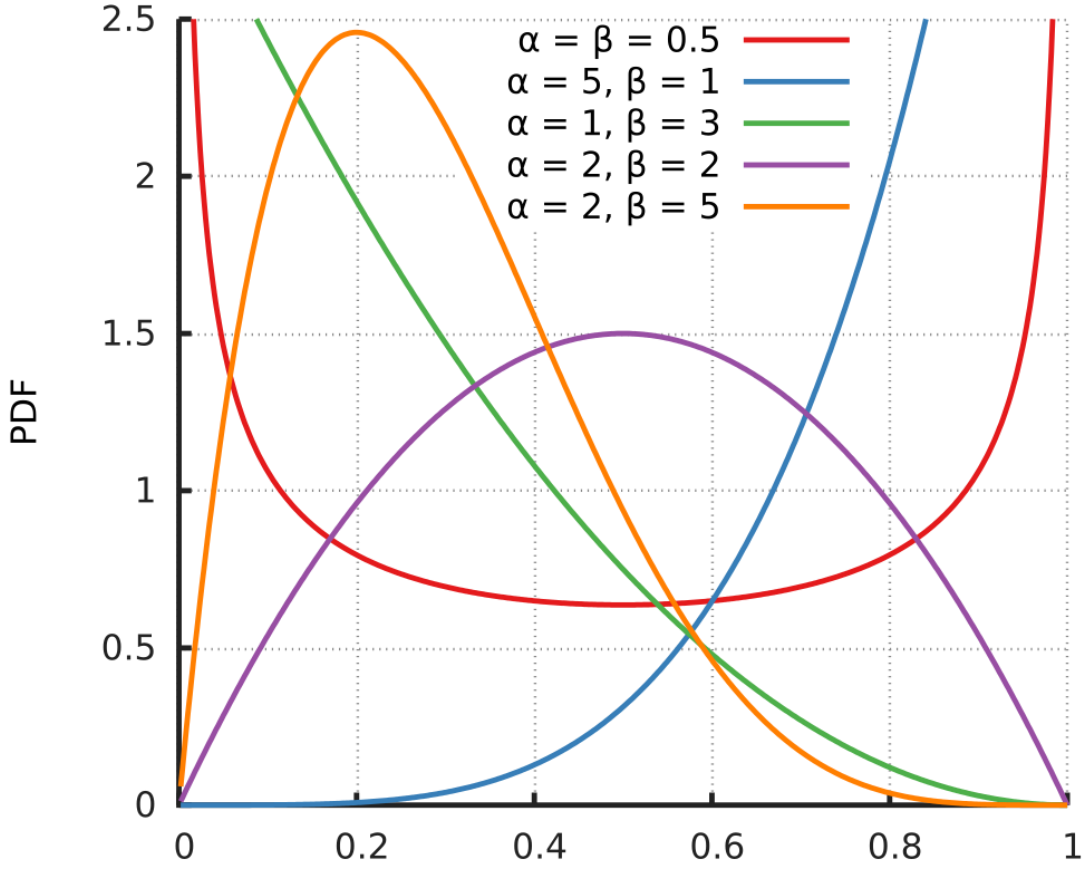
Beta Distribution- Probability Density Function

The parameters **α (alpha)** and **β (beta)** are what *shape* the Beta distribution.

Samples from a Beta distribution **represent** possible probabilities reflecting our **beliefs** about an **unknown binary event's chance of success**.



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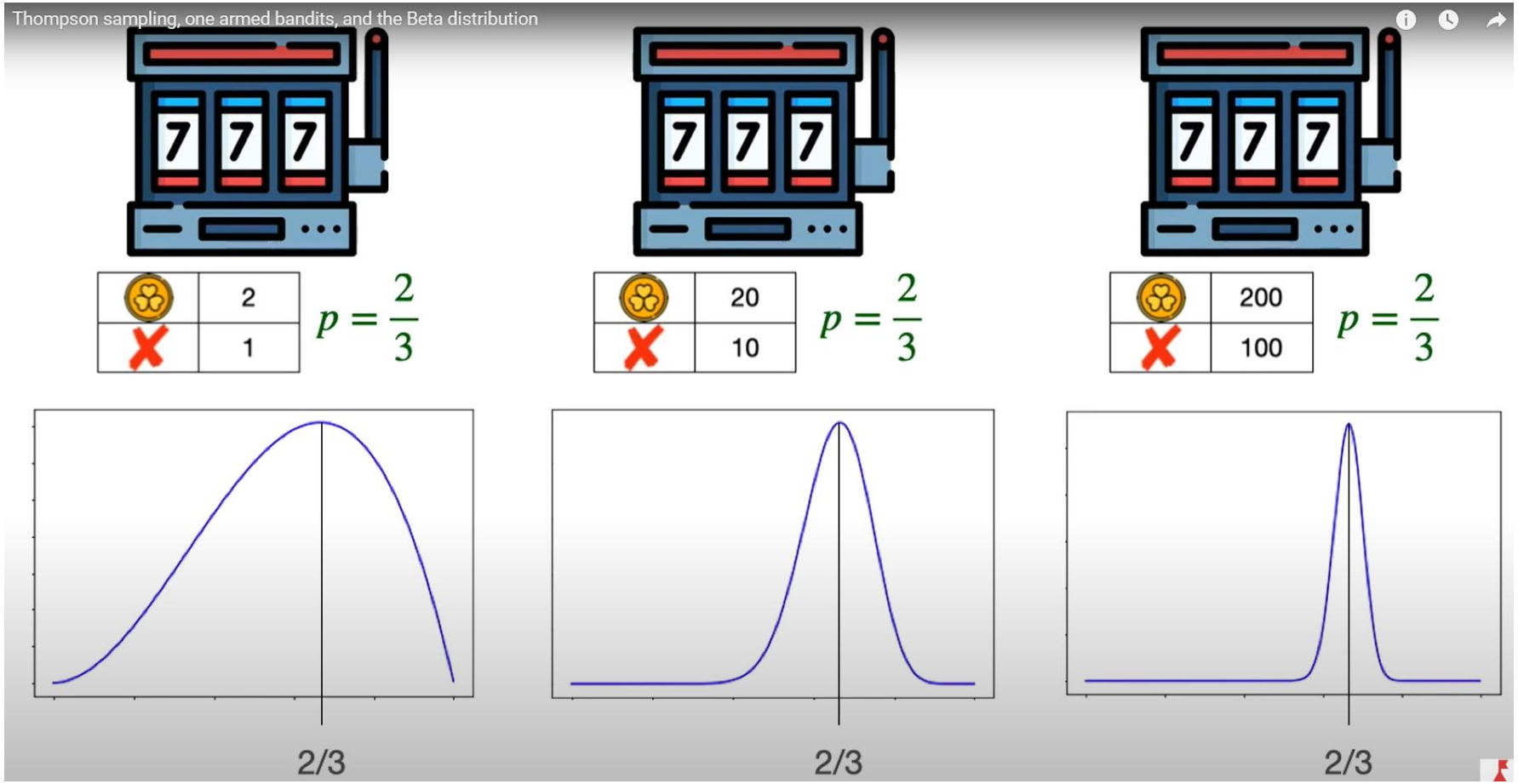
Example Beta Distributions



Src: https://www.youtube.com/watch?v=nkyDGGQ5h60

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Example Beta Distributions

Use for scaling: make AUC 1

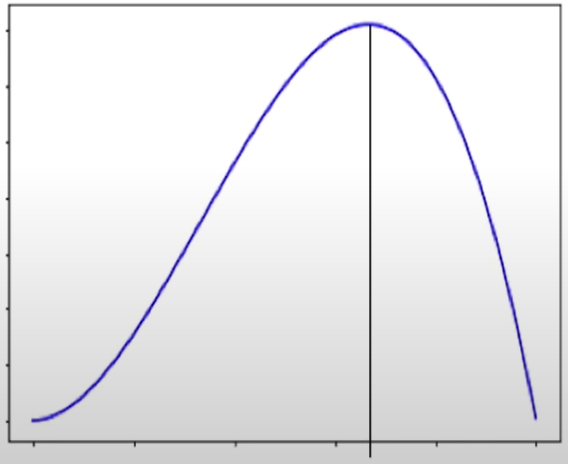
# of success a

# of failure b

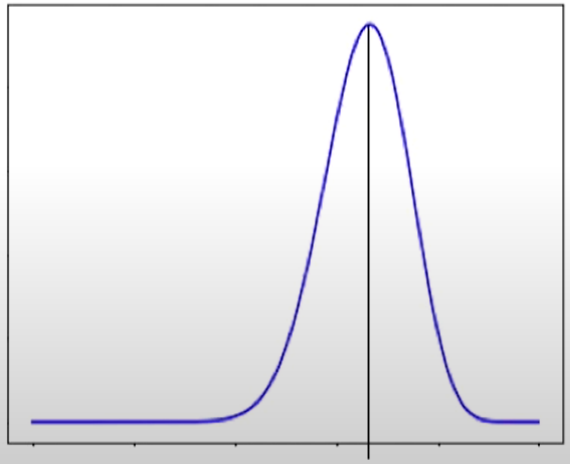
$$f(x) = x^a \cdot (1-x)^b \cdot \boxed{\frac{1}{B(a+1,b+1)}}$$

$$Beta(a+1, b+1)$$

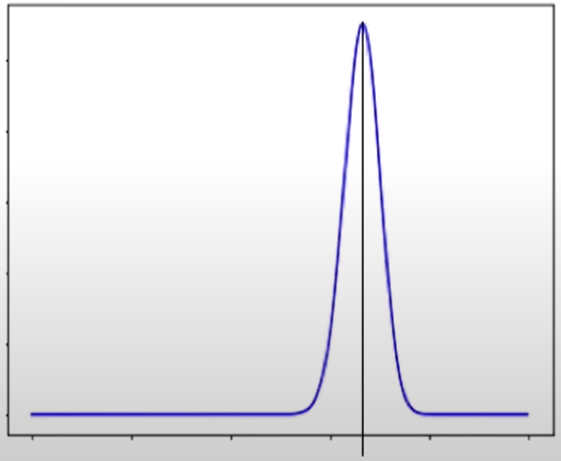
$$f(x) = x^2 \cdot (1-x)^1 \cdot \frac{1}{B(3,2)}$$



$$f(x) = x^{20} \cdot (1-x)^{10} \cdot \frac{1}{B(21,11)}$$



$$f(x) = x^{200} \cdot (1-x)^{100} \cdot \frac{1}{B(201,101)}$$



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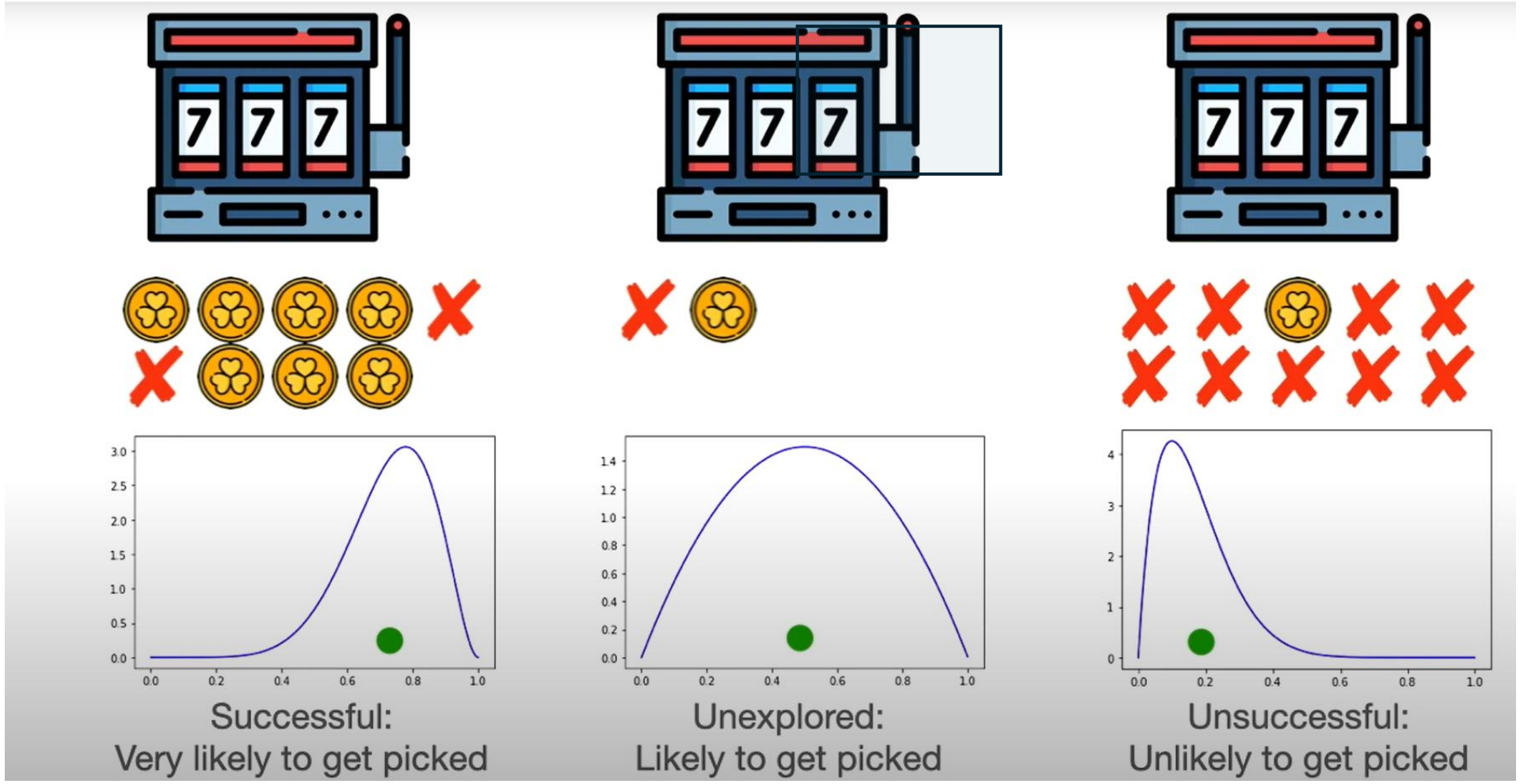
Example Beta Distributions



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How to do this in Python?

Count number of success and failure:

- $\alpha_k \leftarrow \alpha_k + R_t$
- $\beta_k \leftarrow \beta_k + (1 - R_t)$

Where, R_t is the reward (1 or 0)

Use the following simple command (numpy):

Suppose alphas[k] equals α_k, and betas[k] equals β_k;

```
sample = np.random.beta(alphas[k], betas[k])
```

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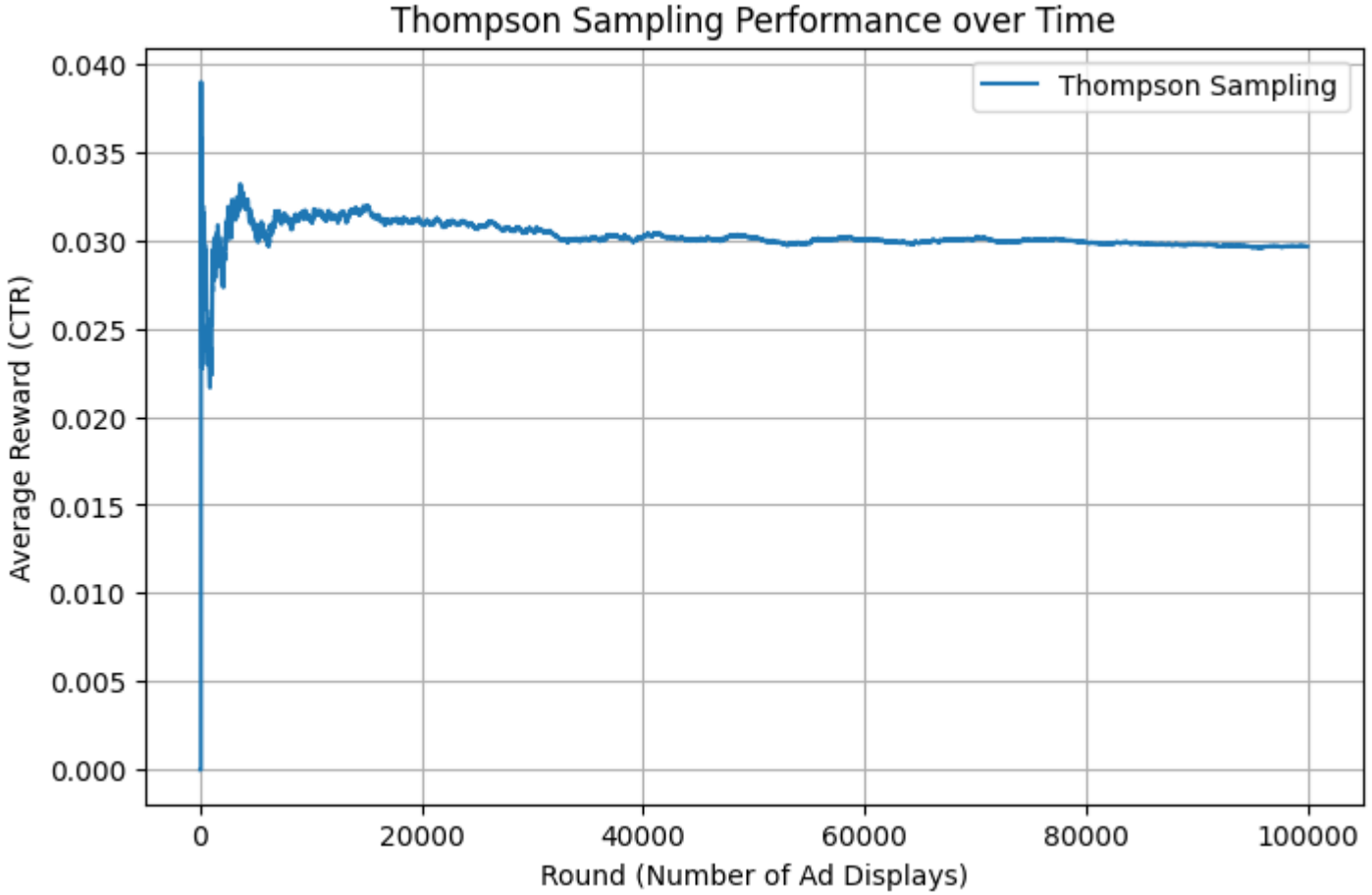
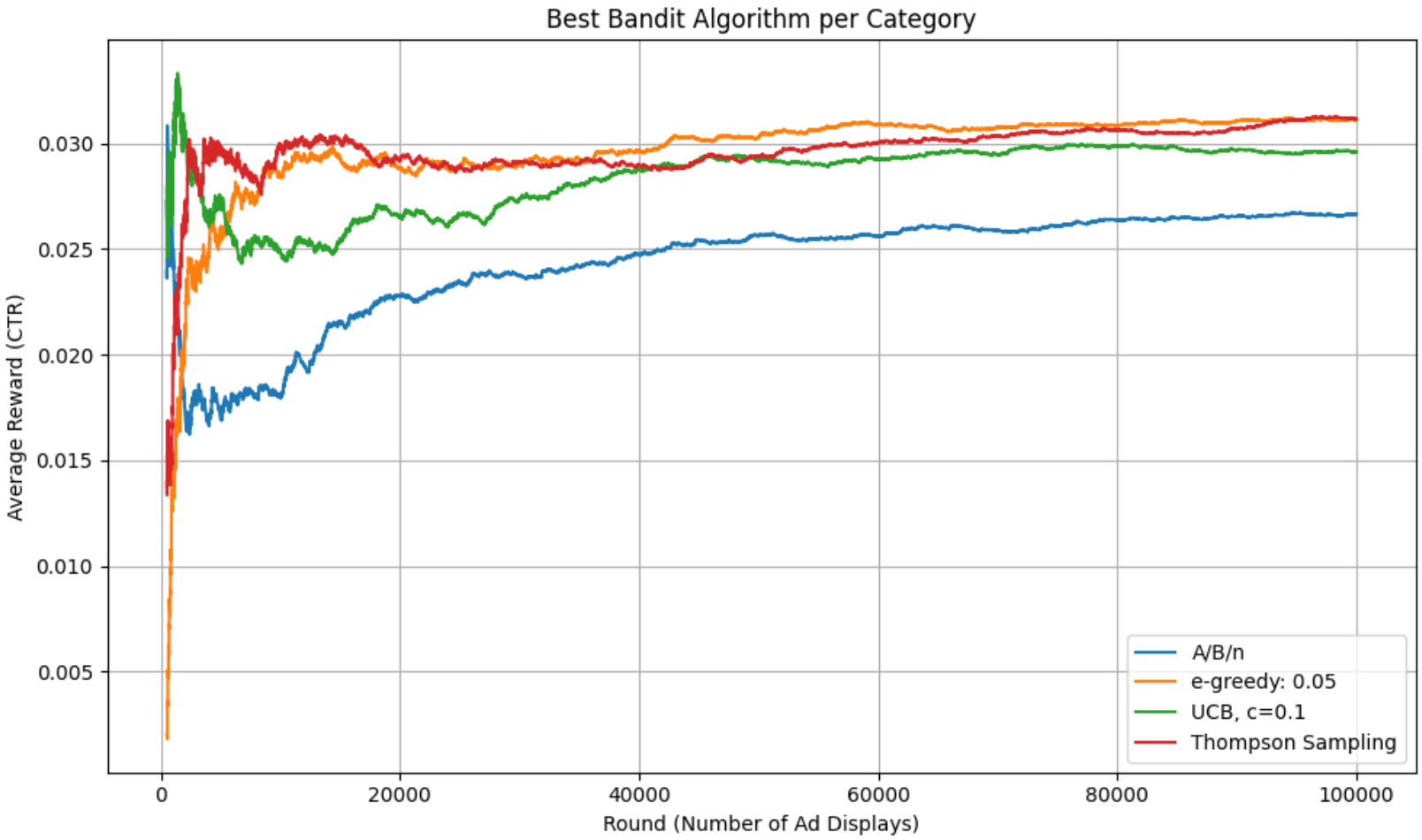
Online Advertising Scenario with Thompson Sampling

- See '**Multi-armed bandits.ipynb**' file in the Teams course page.



Comparing Algorithms

Note that different plots are possible after each run due to stochastic nature of the algorithms



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Next week

Contextual Bandits (CBs)

- CBs allow us to leverage context information (for instance, the user persona, age, gender, location, or previous visits), which means, they play a role in central role in commercial personalization and recommendation applications.